

Report: Frequency De-mixer - 'Unwanted Solo'

Course: EE200 – Signals, Systems, and Networks

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Introduction

In music production, one of the common challenges is isolating and removing unwanted sounds, such as background noise or out-of-sync instruments, to restore or enhance a musical track. In this task, we are provided with an audio track that contains an unwanted instrumental beat, disrupting the intended harmony. The goal is to remove this unwanted component using filtering techniques and restore the original audio as faithfully as possible.

This report outlines the process of designing and implementing a filter to remove the unwanted solo (instrumental beat) from the given audio. The methods used include Fourier Transform-based filtering and the design of a Butterworth bandstop filter to attenuate the unwanted frequency components.

Methodology

1. Input Audio Analysis

The audio file (`before_filter.wav`) contains a mixture of the desired musical signal and unwanted instrumental beats. To begin the filtering process, we first analyzed the time-domain signal and power spectral density (PSD) of the original audio to understand the frequency content.

The time-domain plot provides an overall view of the signal's amplitude over time. The PSD, obtained using the Welch method, shows how the signal's power is distributed across different frequency components.

2. Frequency Domain Filtering

a. Fourier Transform

We used the Fast Fourier Transform (FFT) to convert the time-domain signal into the frequency domain. The frequency spectrum revealed the presence of unwanted frequencies corresponding to the instrumental beats. To remove these unwanted frequencies, we defined a stopband in the frequency range from 1000 Hz to 5000 Hz, where the instrumental beats are most prominent.

By zeroing out the frequencies within this stopband in the FFT domain, we were able to suppress the unwanted components while preserving the rest of the signal. We then applied the Inverse FFT to transform the modified frequency spectrum back into the time domain, resulting in the filtered audio signal.

b. Butterworth Bandstop Filter

In addition to the FFT-based approach, we designed a Butterworth bandstop filter using the `scipy.signal.butter` function. This filter was designed to attenuate the unwanted frequency range while maintaining the rest of the signal. The filter order was set to 10 to achieve a sharp cutoff at the edges of the stopband. The design also included plotting the Bode plots (magnitude and phase responses) to visualize the filter's frequency response and ensure it met the design specifications.

3. Filtered Signal Visualization

a. Time Domain Comparison

We compared the original and filtered audio signals in the time domain to visually assess the effect of the filter. The filtered signal shows the suppression of unwanted instrumental beats while preserving the desired musical components.

b. Frequency Domain Comparison

We also compared the Power Spectral Density (PSD) of the original and filtered signals. The filtered signal shows a significant reduction in power at the frequencies within the stopband (1000 Hz to 5000 Hz), confirming the effectiveness of the filter.

c. Spectrogram Comparison

Finally, we visualized the spectrograms of both the original and filtered signals. The spectrograms show how the frequency content evolves over time, with the filtered spectrogram showing a reduction in the unwanted frequencies in the stopband.

Results

1. Time Domain

The time-domain plot of the original and filtered signals clearly shows that the unwanted instrumental beats have been suppressed. The filtered signal retains the overall structure of the original music while eliminating the disruptive beats.

2. Power Spectral Density (PSD)

The comparison of the original and filtered PSD shows a clear attenuation in the frequency range from 1000 Hz to 5000 Hz, confirming that the filter effectively removed the unwanted frequencies.

3. Spectrogram

The spectrograms of the original and filtered signals provide a visual representation of the frequency content. The filtered signal shows a noticeable reduction in energy in the stopband frequencies, indicating that the filter successfully removed the unwanted instrumental beats.

Discussion

1. Filter Performance

The designed Butterworth bandstop filter successfully removed the unwanted instrumental beats from the audio signal. The Bode plots confirmed that the filter attenuated the desired frequency range effectively, while the time-domain and frequency-domain plots visually validated the results.

The filter order of 10 provided a sharp enough cutoff for this task, but in other scenarios, a higher or lower order might be needed depending on the exact characteristics of the unwanted frequencies.

2. Audio Artifacts

During the filtering process, some audio artifacts, such as minor ringing or phase distortion, may occur, especially near the stopband frequencies. While these artifacts were minimal in this case, they can be mitigated by experimenting with the filter order or by using additional techniques like windowing or Wiener filtering.

3. Limitations and Improvements

The filter was designed to suppress a fixed range of frequencies (1000 Hz to 5000 Hz), which worked well for the given audio. However, for audio tracks with different types of interference, a more adaptive filter might be necessary to handle varying unwanted frequencies.

In future iterations, we could experiment with different filter types (e.g., high-pass or low-pass filters) or employ spectral gating techniques for more precise control over the removal of unwanted sounds.

Conclusion

This task successfully demonstrates the use of both Fourier Transform-based filtering and a Butterworth bandstop filter to remove unwanted instrumental beats from an audio track. The results showed effective suppression of the unwanted frequencies while retaining the quality of the original musical signal.

By analyzing the original and filtered signals in both the time and frequency domains, we ensured that the filtering process was both accurate and efficient. Future work could involve testing more complex filtering techniques for improved performance or handling different types of interference in audio tracks.

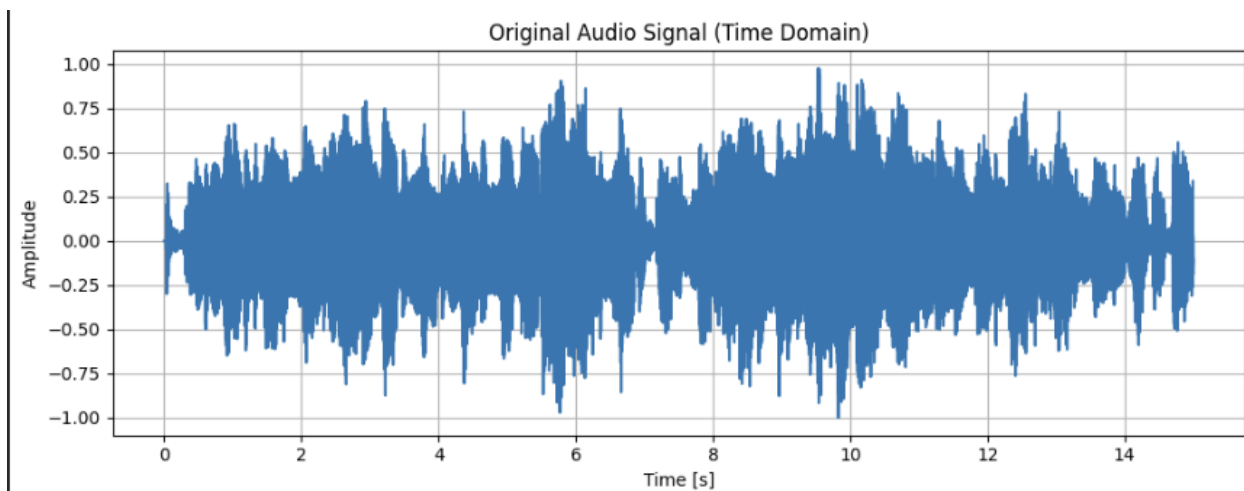


Figure 1: Original Audio Signal (Time Domain)

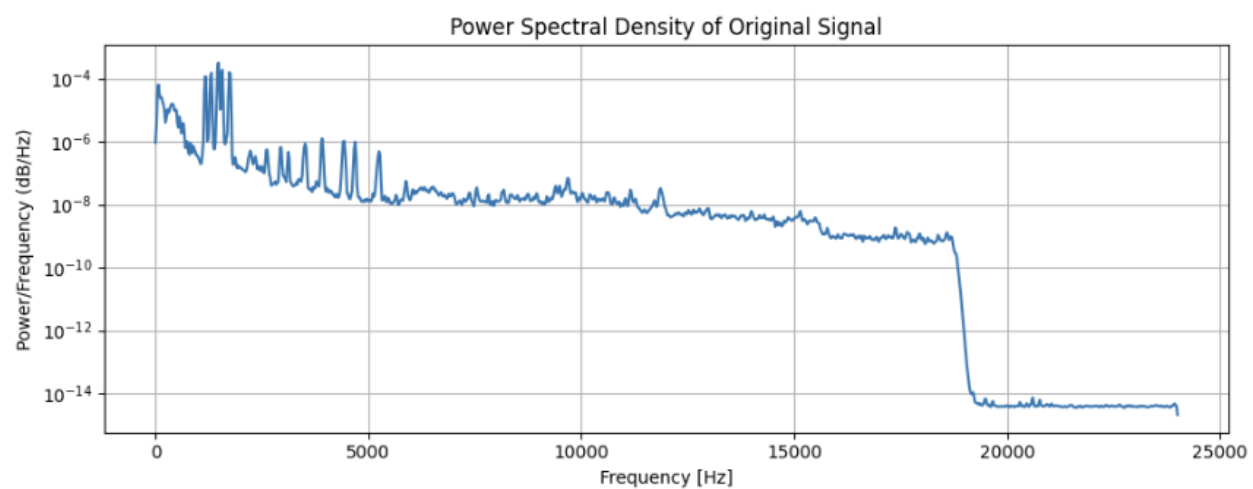


Figure 2: Power Spectral Density of the original signal

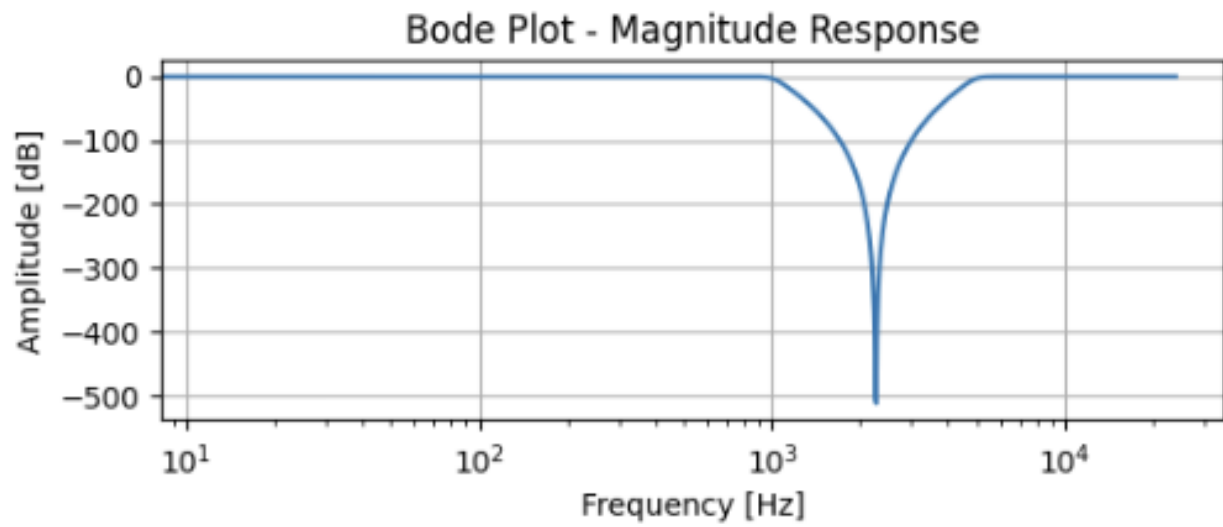


Figure 3: Bode Plot - Magnitude Response

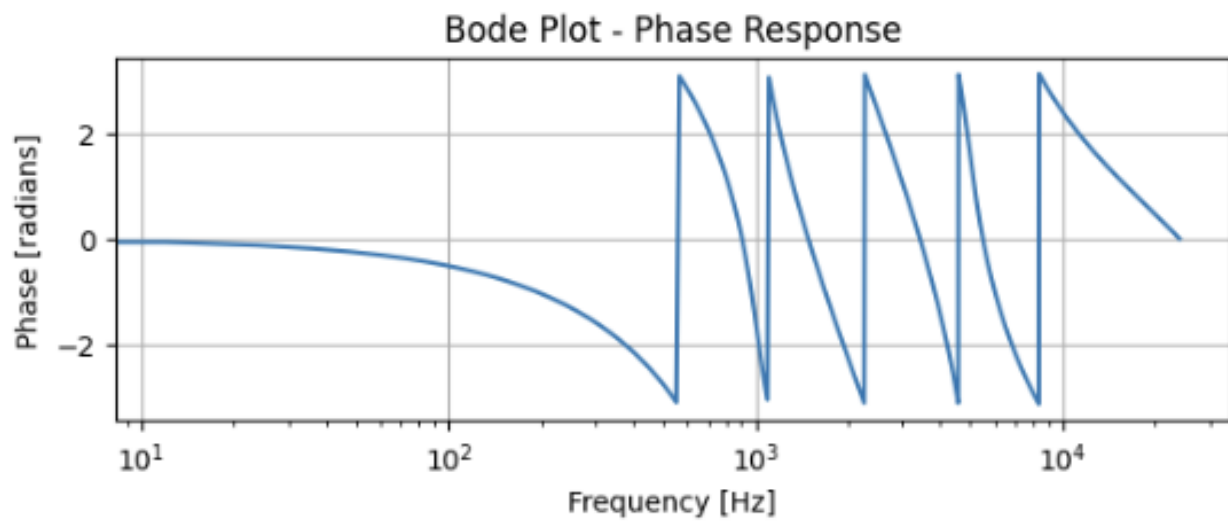


Figure 4: Bode Plot - Phase Response

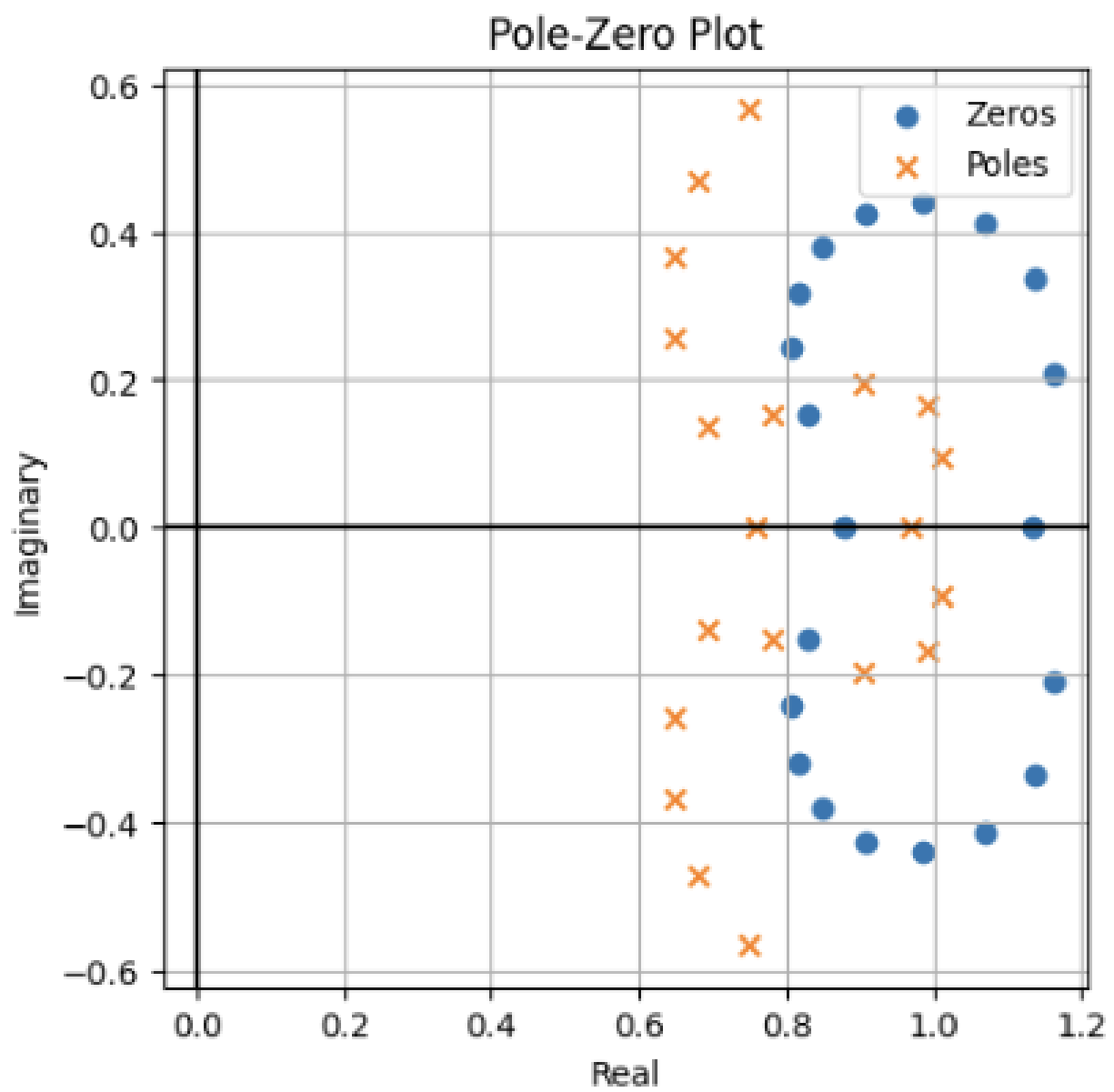


Figure 5: Pole-Zero Plot

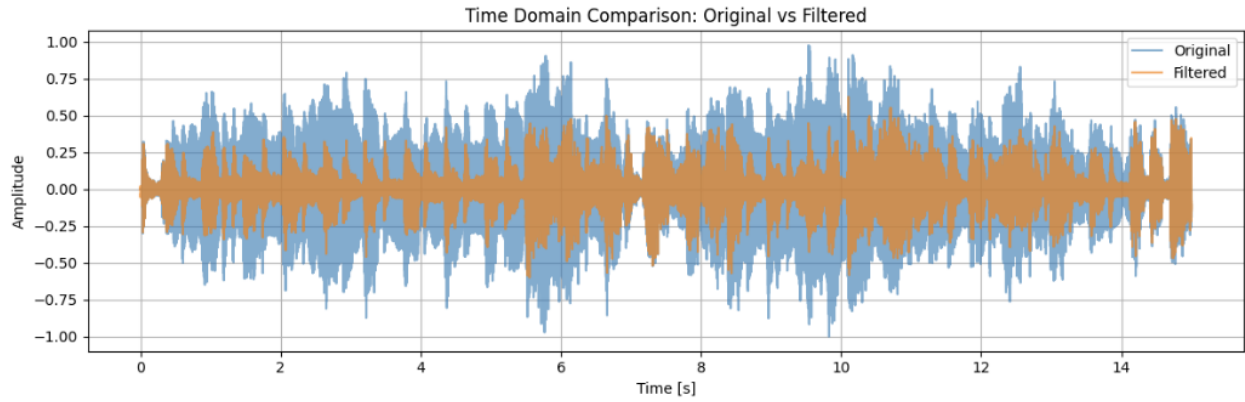


Figure 6: Time Domain Comparison : Original vs Filtered

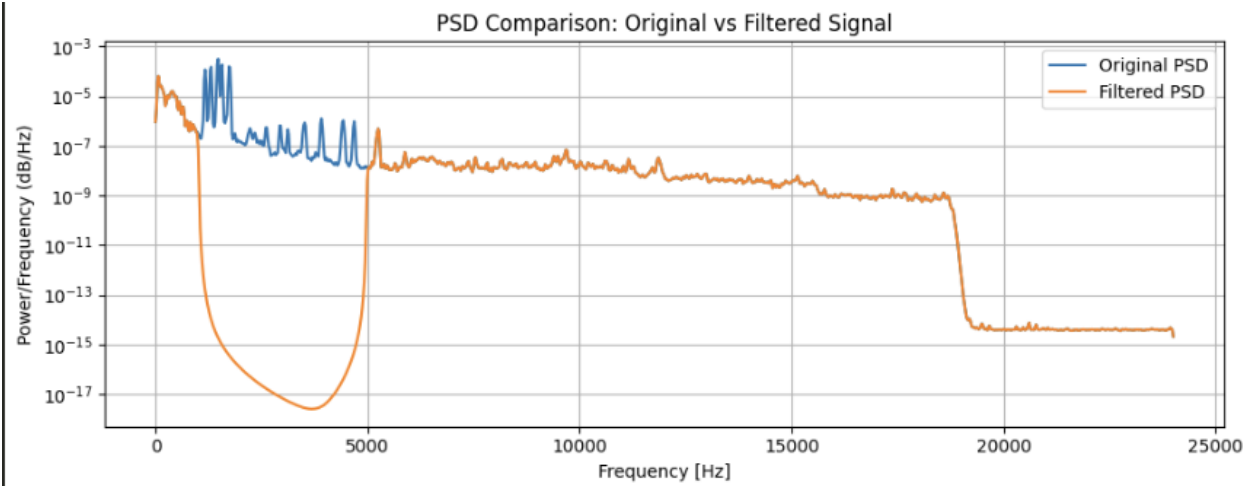


Figure 7: PSD Comparison : Original vs Filtered Signal

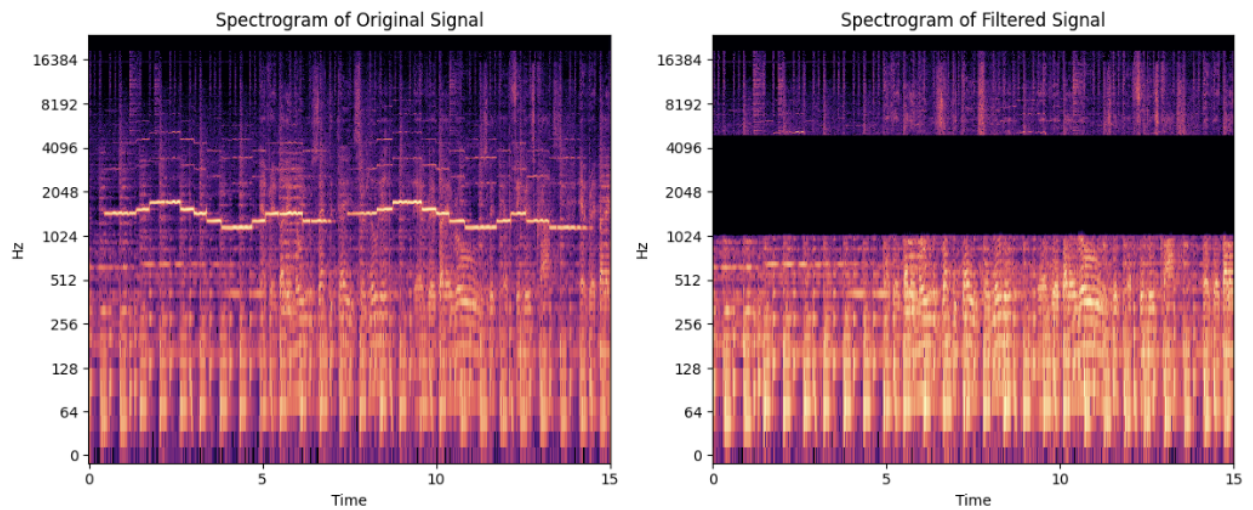


Figure 8: Spectrogram comparison between original and filtered signals